

ROBERT McCRAY

Houston, TX | mccrayrobert26@gmail.com | (713) 399-9544 | Portfolio:
<https://robert-mccray.github.io/robertmccray.github.io/> | Github: github.com/robert-mccray | LinkedIn:
linkedin.com/in/robert-mccray-467613195

SUMMARY

Senior Cloud Engineer & Platform Architect specializing in building governed, multi-cloud platforms and AI infrastructure for strictly regulated environments (SOC2, HIPAA). Highly experienced in translating complex compliance requirements into automated, high-availability architectures using Terraform, Kubernetes, and AWS/Azure. Proven track record of designing "audit-ready" systems that maximize developer velocity while enforcing FinOps, Open Policy Agent (OPA) governance, and zero-trust security at the deterministic commit boundary.

CORE ARCHITECTURE & CLOUD COMPETENCIES

- **Cloud Architecture:** AWS (EKS, Lambda, API Gateway, S3, IAM) • Azure (AKS, ADF, Synapse, VNet) • GCP
- **Platform Engineering & FinOps:** Spotify Backstage (IDP) • Infracost • Open Policy Agent (OPA) • KEDA
- **Security & AI Governance:** Zero-Trust Egress • SOC2/HIPAA • Falco • Cryptographic Ledgers (SHA-256)
- **Data & Distributed Systems:** TimescaleDB • Redis Pub/Sub • Snowflake • Kafka • Medallion Architectures
- **DevOps & Infrastructure:** Kubernetes • Docker • Terraform (IaC) • GitLab CI/CD • GitHub Actions
- **Languages & Observability:** Python • SQL • Bash • FastAPI • Prometheus • Grafana • PagerDuty

KEY PLATFORM ENGINEERING SUITES

The Sentinel IDP & FinOps Engine

- Eliminated manual Jira ticketing bottlenecks by architecting a Spotify Backstage Internal Developer Portal. Enabled 60-second self-service database provisioning via dynamically generated Terraform. Shifted FinOps left by integrating a local LLM to intercept CI/CD Pull Requests, identifying an average of 15% in potential cloud compute savings pre-merge.

Enterprise AI FinOps Gateway & Circuit Breaker

- Architected a stateful FastAPI interceptor mesh to prevent LLM bill shock across distributed applications. Engineered a deterministic token-cost calculator (**tiktoken**) integrated with asynchronous Redis state management to evaluate prompt unit-economics in real-time, automatically triggering HTTP 402 circuit breakers to block negative-ROI or out-of-budget requests before execution.

Zero-Trust AI Governance & Data Lineage Ledger

- Engineered a compliance-ready FastAPI ingestion pipeline that cryptographically hashes (SHA-256)

enterprise documents prior to RAG vectorization. Enforced strict data provenance by storing legal clearance metadata in an immutable SQLite ledger, providing a deterministic verification endpoint to prevent LLM copyright infringement and PII leakage.

Autonomous SRE & Identity

- **Mesh:** Drastically reduced Mean Time To Resolution (MTTR) by deploying an air-gapped Llama-3 SRE Oracle to translate cryptic Falco syscalls into plain-English root cause analyses with executable *kubect1* patches. Replaced static IAM with a Google Zanzibar-inspired Identity Mesh that evaluates behavioral telemetry (IP, time) to execute Just-In-Time (JIT) access revocations.

Distributed Quantitative Trading Platform

- Designed a decoupled, event-driven Docker microservice cluster handling high-frequency time-series data using TimescaleDB and a Redis memory queue. Integrated an air-gapped 7B-parameter Vision AI (LLaVA) for deterministic execution validation, engineering graceful degradation fallbacks to prevent OS-level OOM container crashes.

EXPERIENCE

Cloud Data Platform Architect (Consulting)

Independent Client Engagement - Assisted Living Care | Houston, Tx
December 2025 – Present

Industries: Assisted Living & Medical Care Facilities, AI-Driven Insurance

Problem: Clients operated legacy, siloed operational systems with limited audit visibility, inconsistent data quality, and growing regulatory and contractual pressure, particularly where medical data and AI-driven decisioning intersected. Improved processes for 2 organizations, supporting over 1,000 patients and.

- Architected a governed, multi-tenant Azure data platform using Azure SQL as the authoritative OLTP source, integrating immutable audit ledgers to mathematically guarantee the integrity of medical data and AI decisioning.
- Engineered real-time and batch Medallion data pipelines (Bronze $\rightarrow$ Silver $\rightarrow$ Gold) via Azure Data Factory, extracting legacy OLTP data into ADLS Gen2 for secure downstream analytics.
- Enforced deterministic policy-as-code and data quality contracts at the Silver layer, automating normalization, validation, and explicit reject handling to prevent silent data failures.
- Curated SQL-based semantic layers to demonstrate strict governance at both the storage and database levels, ensuring all AI and reporting models queried validated datasets.
- Designed BI-ready star schemas and Power BI dashboards for operational KPIs, establishing a clear line of sight into data-quality metrics and audit visibility for compliance officers.
- Implemented comprehensive platform observability using Azure Monitor, Log Analytics, and KQL to instantly detect operational regressions, execution failures, and data-quality drift.

Cloud Solutions Architect / Lead Software Engineer

Baltimore Area Sports Health | Baltimore, MD

March 2025 – Present

- Architected and led a HIPAA-compliant, multi-tenant cloud platform on AWS and Azure supporting 25+ organizations, utilizing identity-first architecture and strict RBAC to secure PHI/PII.
- Engineered an immutable Decision-Evidence-Layer to ensure every system decision and data access request was logged, justified, and hash-verifiable for compliance audits.
- Built event-driven, cloud-native ingestion pipelines to process structured and unstructured healthcare data, decoupling high-volume API traffic using message brokers to ensure high availability.
- Implemented active-active Multi-Cloud Redundancy to mitigate single-provider risk, ensuring AWS or Azure regional outages do not halt critical, life-saving clinical care.
- Optimized distributed cloud workloads, leveraging AWS for global disaster recovery and intensive ML pipelines (SageMaker) while utilizing Azure for deep FHIR/DICOM integration and Entra ID identity management.
- Designed least-privilege access paths using Cognito and OAuth2, enforcing zero-trust boundaries across all microservices and user interactions.
- Led NoSQL schema design in MongoDB Atlas, modeling document structures around access patterns and strict tenancy boundaries, integrated with PostgreSQL and Snowflake in a hybrid transactional/analytical architecture.
- Embedded platform-level observability using Prometheus and Grafana, establishing SLO/SLI metrics, incident correlation, and error budgets to support continuous operational monitoring.

Independent Cloud & Data Consultant

Houston, TX

Jul 2024 - March 2025

Industries: Education & Tutoring Services, Private Medical Practice

Problem: Clients relied on manual processes and fragmented systems for scheduling, billing, reporting, and student/patient data management. Data inconsistencies, lack of lineage, and limited observability created operational risk and constrained growth—particularly for medical practitioners handling sensitive data for 3 separate organizations.

- Designed and validated governed cloud architectures for regulated systems, shifting from manual infrastructure to Infrastructure-as-Code (Terraform) to ensure repeatable, audit-ready deployments.
- Implemented compliance-aware data protection strategies, including modeling Write-Once-Read-Many (WORM) storage principles and strict namespace isolation to protect sensitive records from tampering.
- Enforced clear data ownership and lineage tracking across distributed databases, preventing reporting discrepancies and eliminating silent data corruption.
- Produced detailed target-state architecture diagrams, governance artifacts, and analytics prototypes to align stakeholders on security-by-design principles prior to downstream implementation.

Cloud Systems Engineer

Keystone Advisors | Houston, TX

March 2024 – June 2024

- Designed and orchestrated Azure Data Factory pipelines for CMS-regulated healthcare reporting for 100+ vendors

- Integrated Azure Health Data Services to normalize HL7/FHIR data at scale.
- Tuned Azure Synapse ETL performance, reducing execution times .
- Implemented row-level security (RLS) in Power BI dashboards for executive and compliance stakeholders.
- Automated CMS API integrations, reducing failure rates through resilient retry and schema validation logic.

Cloud Systems Engineer

Bel BeautEssentials | Houston, TX

October 2022 – March 2024

- Managed AWS IAM roles and access controls, reducing over-permissioned accounts
- Supported backend data storage and application services using MongoDB-based NoSQL patterns for operational data and user workflows
- Assisted with data modeling and collection structure design to support evolving application requirements
- Managed AWS IAM roles and access controls, reducing over-permissioned accounts
- Supported data movement between application data stores and downstream reporting systems
- Contributed to automation and deployment workflows for cloud-based services

EDUCATION & CERTIFICATIONS

- **Bachelors of Liberal Arts and Sciences** - Indiana University (May 2018)
- **DevOps Professional** - PagerDuty (Incident response, observability, reliability)
- **Docker Foundations Professional Certificate** - Docker
- **Introduction to Finops** - FinOps Foundation
- **Cloud Strategy Professional Certificate**
- **Hands-on Data Science & AI for Healthcare** (Healthcare analytics, ML pipelines, PHI gov)